

Design of a 4-bit Digital to Analog Converter

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Abstract— This project (DAC), takes 64-bit audio from a digital source and outputs the same audio, but now in four-bit. In the project we utilize an Arduino, an in-house-built level shifter, and a custom-built PCB ordered from Osh Park to deliver the four-bit audio to a speaker.

Keywords—DAC

I. INTRODUCTION

A DAC, or a digital-to-analog converter, takes in digital data and transforms it into an analog audio signal. These are used in nearly any device with digital sound such as CDs, game consoles, laptops, and phones. This digital format is changed into an analog format that is capable of being output. [1][2].

Our DAC uses voltage sources with a binary weight. There exist other DACs that use a current mode instead of voltage, but with the same binary weight. Our DAC only has a 4-bit input, but other designs can convert 8- and 16-bit inputs. Another type of DAC is the R-2R resistor ladder network. [3][4].

Our design is simple and relatively cheap, making it the perfect project for educational purposes. The whole process gives a solid introduction to CAD along with PCB design, while also building a foundation on the functionality of op-amps. All of this allows for a great hands-on experience for beginning engineers.

The following section will discuss the exact 4-bit DAC design that was used along with computer simulations backing the decision. The third section covers PCB design using Fusion 360. The fourth section explains the soldering and testing process of the PCB board and section five goes more in depth about the applications. Finally, section six rounds out with a conclusion.

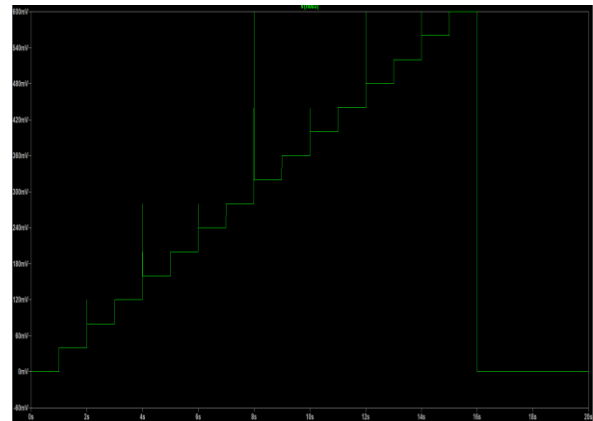


Figure 1: DAC Simulation Output

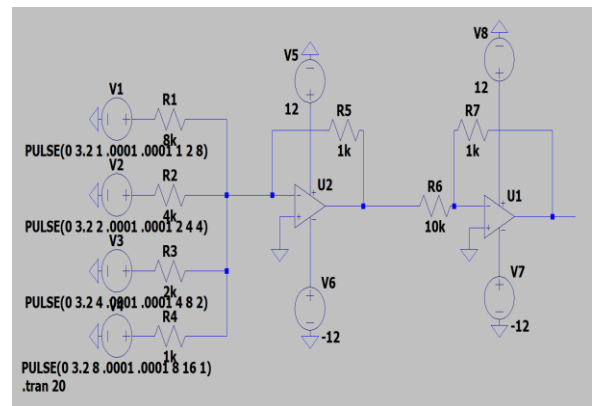


Figure 2: 4-bit DAC Schematic (LTspice)

II. 4-BIT DAC DESIGN AND SIMULATION

Using the concept of binary numbers, we were able to design a summing amplifier with 4-bit values. This allowed us to create 16 distinct levels of voltage. However, the magnitude of these voltages would be too large and negative. Which is why an inverting amplifier with a fractional gain would flip the outputs and make them $1/10^{\text{th}}$ of their original value and positive. Leaving us with voltage levels favorable to use in the provided audio jack.

The SPICE schematic (Figure 1) took all of this into account and drew up a design with 4 voltage sources with their respective resistor value to represent bit values and 2 amplifiers to provide the 16 levels of voltage then its more applicable version of those 16 levels. The components used were thought-out to ensure the design functioned properly. The operational amplifier was supplied, but the resistor values were chosen based on availability and the pros and cons of large or small valued resistors.

Finally, the simulated output of this schematic proved to be functional as all the combinations of the 4-bit values worked out to provide 16 levels of voltage and the inverting amplifier with fraction gain successfully altered the values to the range and sign of values we were looking for (Figure 2).

III. PCB DESIGN

To power the circuit, we created a three pin header for $\pm 12V$ and ground. We created a 5 pin header for the four DAC inputs and ground. 7 copper resistor pad pair mounts for respective SMT resistors. 8 appropriately connected copper pads for SMT LM1458 op amp and another three copper pads for an audio jack.

From the schematic, we generated a 2-layer PCB board layout as shown in Figure 3. We used FUSION's auto-router as well as manual routing to complete connections to every input and output. We used mostly the default FUSION design or rules file, except for copper width, which was changed to 10 mil, drill size which was changed to 20 mil, and board outline clearance which was changed to 40mil. We first began with setting up the platform where our components would sit. This consisted of creating a 1" x 1" square and drilling 4 125 mil diameter holes in the corners at the coordinates (100, 100), (100, 900), (900,100), and (900, 900) mil to be able to mount our PCB to our 3D printed chassis.

The rest of the board was for our components, the first of which we decided to place the audio jack since it had to be placed specifically so that part of it hangs off the edge. Then certain areas were allocated to hold specific components. One section for resistors, another for the amplifier.

The final step was to route the design, which began with using the autoroute tool. Any unrouted connections used vias to under already routed wires or we manually created routes to allow for more connections. Then we added the text labels for ease of assembly and checked design rules which left us with the completed PCB Board Layout (Figure 3).

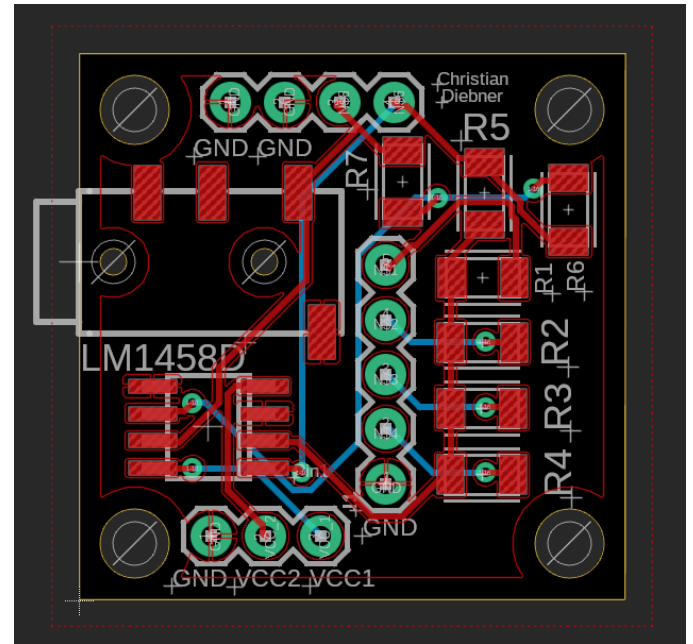


Figure 3: DAC PCB layout

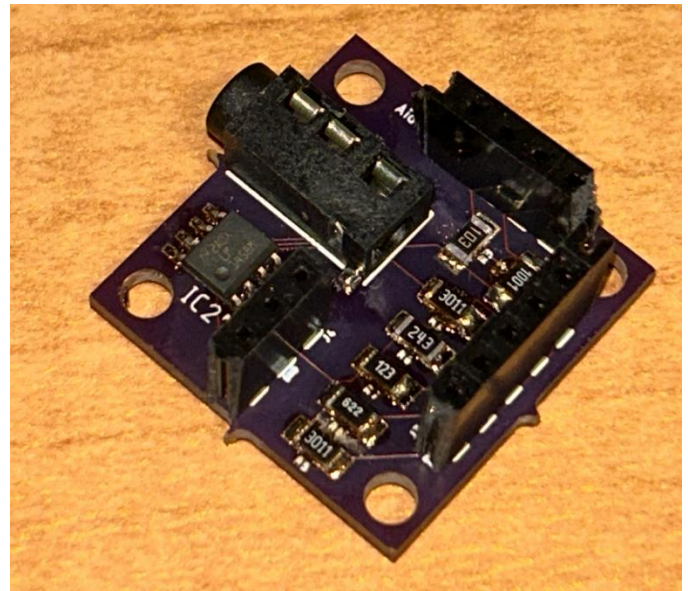


Figure 4: PCB Board Layout of DAC Circuit

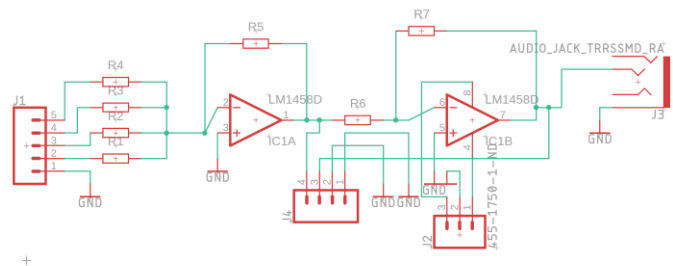
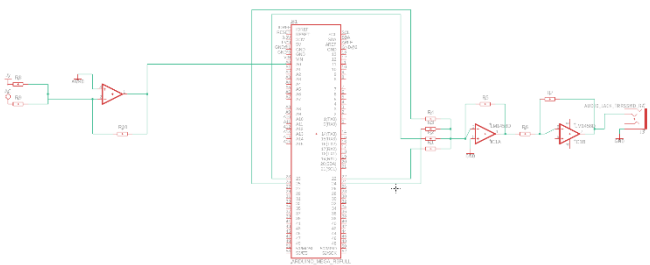


Figure 5:PCD schematic

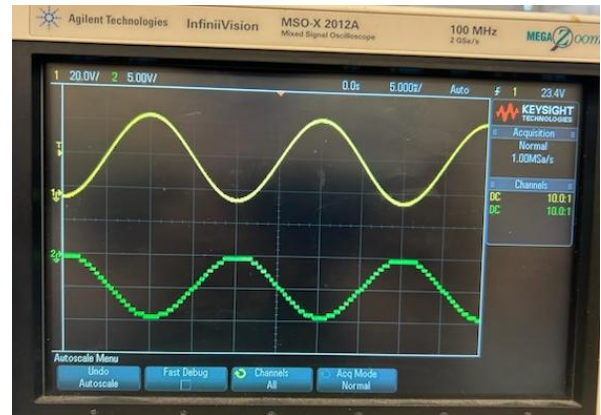
We ordered the PCBs from OshPark, the rest of the electrical components came from the professor-constructed lab kit. OshPark, a low-cost PCB fabrication company, only charges \$5 per square inch for two-layer boards and also provides three boards per purchase. We assume the price for just 1 of these fully assembled boards to be less than 6 dollars.

We powered the board with a $\pm 12\text{V}$ supply and checked for proper operation by checking the outputs of each amplifier using a circuit that would cycle through each 16 levels of the binary inputs (figure 5). We then compared these values to table generated from the LTspice simulation. The observed and the expected had a low margin of error so we concluded that the PCB functioned as it was supposed to.



V. 4-BIT DAC APPLICATION

would produce its own digital wave which is similar to its original analog wave (figure 7).



The way we used this was by having the AC source being music playing from a laptop. The resulting wave was put into a speaker which led to a distorted version of the original sound. This can be observed in the YouTube video provided, a part also should a comparison of the waves.

The ability that our DAC has to convert 64-bit audio from a digital source to four-bit audio may be in itself not extremely useful, but the ideas and concepts behind it are. Figuring out and being able to convert signals to fit into other equipment is extremely useful when trying to design a computer on which not every device can hold the same voltage or current. It can also be useful when trying to make smaller, larger or simply altered forms of that equipment.

John, Diego, Christian would like to thank Jeremy Blum for creating excellent YouTube tutorials for FUSION [5]. Include any other acknowledgements you would like

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